SESSION 2 • CHAPTER 2

AIOps Reference Architecture

From Core Components to OCI Implementation

Core Components

Data & ML Layers

Automation & Feedback

OCI Architecture

Session 2 Overview

What we will cover in this chapter

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AIOps Reference Architecture Overview

The big picture — layers, flows and integration points

02

Core Components of AIOps Platforms

Observe, Engage, Act — the three-pillar framework

03

Data Collection & Ingestion Layer

Metrics, logs, traces, events — unifying the data plane

04

Machine Learning & Analytics Layer

Anomaly detection, correlation, root-cause, forecasting

05

Automation & Orchestration Layer

Automated remediation, self-healing, runbooks

06

Feedback Loops & Continuous Learning

Closing the loop — how AIOps improves over time

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OCI Reference Architecture for AIOps

Oracle Cloud Infrastructure O&M services mapped to AIOps

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OCI Service Deep Dives

APM, Logging Analytics, Ops Insights, Stack Monitoring

AIOps Reference Architecture — The Big Picture

A layered model connecting data sources to intelligent operations

DATA SOURCES

Infrastructure | Apps | Cloud | ITSM | Network

DATA INGESTION

Streaming · Batch · APIs · Agents · OpenTelemetry

ML & ANALYTICS

Anomaly Detection · Correlation · RCA · Forecasting

AUTOMATION ENGINE

Remediation · Orchestration · Self-Healing · Runbooks

BUSINESS OUTCOMES

Reduced MTTR · Lower Costs · Service Reliability · SLA

FEEDBACK LOOP

The Three Core Pillars of AIOps

Observe → Engage → Act: The foundational framework for AIOps platforms

OBSERVE
Collect & Detect

- Continuous monitoring of all IT telemetry
- Metrics, logs, traces, and events ingestion
- Anomaly detection using ML baselines
- Real-time alerting for threshold violations

Tools: Monitoring · APM · Log Analytics · Stack Monitoring

ENGAGE
Analyze & Prioritize

- Incident management and triage automation
- Root cause analysis via AI/ML models
- ITSM integration — auto-ticket creation
- SLA and capacity management oversight
- Change impact and risk assessment

Tools: ITSM · ServiceNow · CMDB · Change Management

ACT
Automate & Heal

- Automated remediation via predefined playbooks
- Self-healing: auto-scale, reroute, restart
- Runbook execution with step-by-step guidance
- Feedback to ML models for continuous learning
- Disaster recovery automation triggers

Tools: Automation · OCI Functions · OCI Connector Hub

Data Collection & Ingestion Layer

The foundation — gathering signals from across the IT estate

Metrics

CPU, memory, latency, throughput, error rates — time-series performance data from infra & apps

Logs

Textual event records from applications, OS, databases, security tools — key for root cause

Traces

End-to-end distributed transaction flows across microservices using OpenTelemetry / OpenTracing

Events

CI/CD deployments, config changes, alerts, ITSM tickets, cloud API state changes

Ingestion Mechanisms

Agent-Based

Lightweight agents on hosts push metrics & logs to central platform (OCI Management Agent, Fluentd)

API Pull

Platform polls REST/SNMP endpoints of cloud services, databases, network devices on schedule

Streaming

Real-time event streams via Kafka, OCI Streaming — millisecond latency for high-velocity data

OpenTelemetry

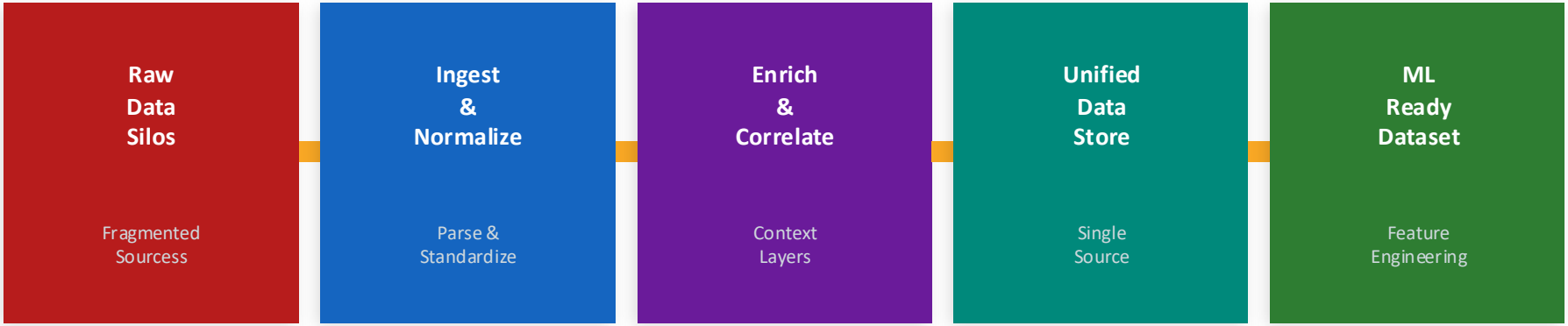
CNCF standard SDK for auto-instrumenting apps — unified traces, metrics, logs collection

Service Connectors

OCI Service Connector Hub orchestrates data movement between OCI services and third-party tools

Data Normalization & Unified Data Platform

Breaking silos — creating a single source of truth for AIOps analytics



Why Normalization Matters

- Different tools use different timestamp formats, field names, and severity levels
- ML algorithms require consistent, clean, structured input for accurate results
- Correlation across systems is impossible without a common data schema
- "Garbage in, garbage out" — poor data quality is the #1 cause of AIOps failure

Data Challenges

10x	More telemetry from microservices vs monoliths
87%	Enterprises run hybrid/multi-cloud generating fragmented data
60%+	Of traditional alerts are noise or duplicate signals
~5TB	Daily log volume in a typical large enterprise environment

Source: Netdata AIOps Architecture | Mordor Intelligence | FlowHunt AIOps Guide | https://www.netdata.cloud/academy/aiops/

ML & Analytics Layer

The intelligence engine that transforms raw data into actionable insights

Anomaly Detection

Dynamic ML baselines replace static thresholds. Identifies deviations in metrics, logs, and traces with contextual awareness of time-of-day, seasonality, and deployment changes.

Event Correlation

Groups related alerts across systems into a single meaningful incident. Reduces alert noise by up to 99% by linking symptoms to root causes across infrastructure layers.

Root Cause Analysis

AI traces the causal chain from symptoms to origin. Correlates recent changes (deployments, config) with anomaly onset to pinpoint which component is actually failing.

Predictive Analytics

Time-series forecasting predicts resource exhaustion, capacity shortfalls, and failure patterns 7–30 days ahead, enabling proactive planning before impact occurs.

Pattern Recognition

Unsupervised learning discovers recurring incident patterns and seasonal behaviors in historical data, building institutional knowledge for faster future resolution.

Topology Awareness

ML models understand service dependency maps (CMDB) to assess blast radius, prioritize alerts by business impact, and route issues to the right teams automatically.

Anomaly Detection: From Static to Dynamic

How ML replaces brittle thresholds with contextual intelligence

Traditional: Static Thresholds

- Fixed rule: 'Alert if CPU > 85%'
- Fails to adapt to time-of-day patterns
- Ignores context (deployment, traffic spike)
- Generates thousands of false positives
- Misses slow-burn anomalies below threshold
- Requires constant manual tuning by humans

VS

AIOps: Dynamic ML Baselines

- Learns normal behavior patterns over time
- Adapts baselines per time-of-day and weekday
- Incorporates deployment events as context
- Correlates anomalies across multiple signals
- Detects subtle degradation before user impact
- Self-updates baselines as system evolves

Common ML Algorithms Used in AIOps Anomaly Detection:

Isolation Forest	LSTM Neural Networks	ARIMA / SARIMA	k-Means Clustering	Random Forest	Bayesian Networks

Automation & Orchestration Layer

Closing the gap from insight to action — the Act pillar

Detect → Correlate → Prioritize → Select Playbook → Auto-Remediate → Verify & Learn

Automated Remediation

Predefined scripts auto-resolve known issues — restart service, clear queue, re-route traffic — without human intervention.

Self-Healing Systems

Infrastructure auto-scales, re-provisions, or fails over when anomalies are detected. Zero-incident enterprise (ZIF.AI model).

Runbook Automation

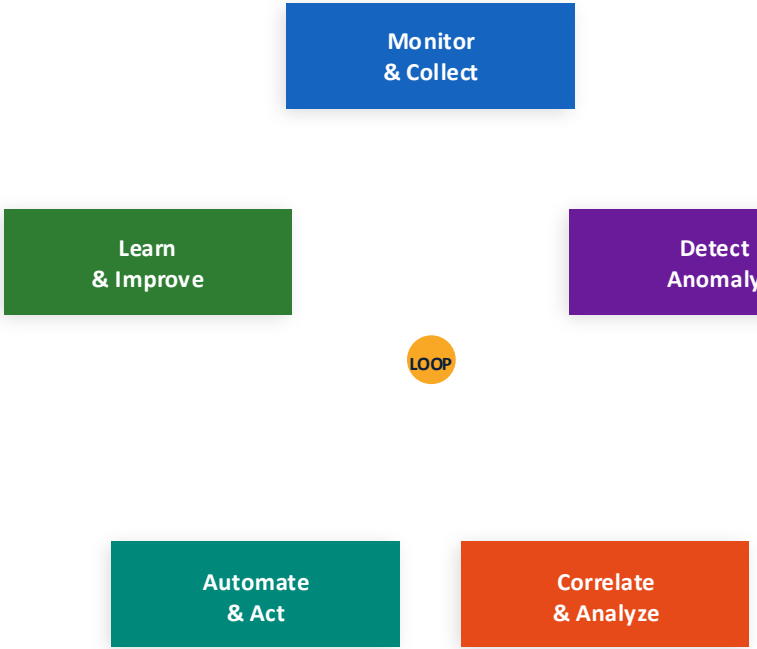
Step-by-step guided resolution playbooks from ITSM knowledge base. AIOps selects the right runbook automatically for each incident.

ITSM Orchestration

AIOps integrates with ServiceNow, Jira, or PagerDuty to auto-create, assign, and route tickets based on incident type and severity.

Feedback Loops & Continuous Learning

How AIOps gets smarter — the self-improving operations cycle



What Continuous Learning Enables

↓ False Positives	ML models refine anomaly thresholds based on confirmed vs. noise alerts over time
↑ Accuracy	Root cause models become more precise as they learn from historical incident resolutions
↑ Automation	New playbooks are generated from successful manual remediations and added to the library
↓ MTTR	Pattern recognition accelerates diagnosis for recurring incident types and failure modes
↑ Coverage	Models expand to new services and infrastructure as they observe new telemetry patterns

Data, ML & Automation: Working Together

How the three AIOps layers interact in a closed-loop system

DATA LAYER

- Collects raw telemetry from 100s of sources
- Normalizes heterogeneous formats
- Creates unified observability schema
- Ensures real-time + historical data availability
- Provides feature store for ML models

OUTPUTS

- Structured Metrics
- Indexed Logs
- Trace Graphs
- Event Streams

ML LAYER

- Consumes unified data from data layer
- Applies supervised + unsupervised learning
- Generates anomaly scores and incident clusters
- Predicts future states from trend analysis
- Feeds confidence scores to automation layer

OUTPUTS

- Anomaly Scores
- Correlated Incidents
- RCA Reports
- Predictions

AUTOMATION LAYER

- Receives prioritized incidents from ML layer
- Selects appropriate remediation playbook
- Executes automated or assisted fix actions
- Captures remediation outcome as training data
- Updates feedback loop for model retraining

OUTPUTS

- Remediation Actions
- ITSM Tickets
- Runbook Executions
- Feedback Data

◄ FEEDBACK LOOP — Automation outcomes re-train ML models; ML insights guide data collection priorities ►

OCI Reference Architecture

for AIOps

Oracle Cloud Infrastructure Observability & Management Platform

What OCI Provides:

- Full-stack observability — from bare metal to application layer
- Native ML-powered log analytics with anomaly detection and pattern recognition
- Application Performance Monitoring (APM) with distributed tracing
- Operations Insights for capacity planning and SQL performance using 25-month historical data
- Service Connector Hub for orchestrating data movement between OCI services
- Stack Monitoring for proactive infrastructure and application health checks
- Integration with open standards: OpenTelemetry, Prometheus, Grafana, PagerDuty, Slack

OCI Observability & Management — Service Landscape

The full suite of OCI services that enable AIOps capabilities

OCI Monitoring

Metrics

Collect & query metrics; configure alarms with MQL — Monitoring Query Language

OCI Logging

Logs

Centralized log management — audit, VCN flow, service logs with full lifecycle mgmt

OCI Logging Analytics

AI/ML Logs

ML-powered log analysis — pattern recognition, anomaly detection, correlation

OCI APM

Tracing

Distributed tracing + end-user monitoring; OpenTracing & OpenMetrics compatible

OCI Ops Insights

Capacity & SQL

ML-based capacity forecasting using 25-month history; SQL trend analysis

OCI Stack Monitoring

App Stacks

Proactive monitoring of full application stack including app servers & databases

OCI Database Management

Database Ops

Fleet monitoring, Performance Hub, real-time SQL monitoring for Oracle DBs

Service Connector Hub

Data Routing

Orchestrates data movement between OCI services and third-party SIEM tools

OCI Notifications

Alerting

Pub/sub alerting via email, Slack, PagerDuty, OCI Functions

Source: Oracle OCI Observability Docs | https://docs.oracle.com/en-us/iaas/Content/cloud-adoption-framework/monitoring-and-observability.htm | https://blogs.oracle.com/observability/

OCI AIOps Reference Architecture Diagram

How OCI O&M services map to the AIOps Observe → Engage → Act model

DATA SOURCES: OCI Compute · OCI Network · OCI Database · Applications · On-Premises · Kubernetes / Containers

COLLECTION LAYER: OCI Management Agent · OpenTelemetry SDK · Fluentd · OCI Service Connector Hub

OBSERVE	ENGAGE	ACT
OCI Monitoring (MQL Alarms)	OCI Logging Analytics (ML)	OCI Functions (Runbooks)
OCI Logging (Log Events)	OCI Ops Insights	OCI Notifications (PagerDuty)
OCI APM (Traces)	DB Management (SQL Trends)	ITSM Integration
OCI Stack Monitoring	Events & Notifications	Auto-Scale (OCI Policies)

◄ **FEEDBACK LOOP: OCI Logging Analytics ML models learn from each incident resolution cycle** ►

INTEGRATION: Grafana · PagerDuty · Slack · OpenTracing · CloudEvents (CNCF) · Enterprise Manager · Third-Party SIEM

Compartment & IAM governance applies across all OCI O&M services for security and access control.

Source: Oracle OCI Observability Reference Architecture | https://docs.oracle.com/en/solutions/oracle-cloud-observability/index.html

OCI Logging Analytics — ML-Powered Log Intelligence

Deep dive: AIOps capabilities within OCI's premier log analysis service

Ingest & Index

Collect logs from OCI services, on-prem, cloud-native apps. Supports 300+ out-of-box log source types with field enrichment.

ML Pattern Detection

Curated AI/ML models automatically detect anomalies, cluster log patterns, and surface unusual sequences without manual rules.

Correlation Engine

Correlate log activity across multiple systems — link an app error to a DB slowdown and a network event as one root cause chain.

Visualization

Interactive dashboards with log explorer, trend charts, heatmaps, and topology views for rapid incident investigation.

Alerting & Actions

Configure log-based alarms to trigger OCI Notifications, Functions, or Service Connector Hub pipelines for automated response.

Key AIOps Use Cases

Security SIEM	Feed audit logs into SIEM for threat detection and compliance monitoring
Root Cause	Search and correlate logs across app + infra layers to identify incident origin
Capacity Audit	Track resource utilization logs to identify over-provisioned or idle systems
DevOps Insight	Monitor deployment logs to correlate releases with performance degradation
Compliance	Retain, search, and report on audit logs for regulatory requirements

Source: OCI Logging Analytics Docs | https://docs.oracle.com/en-us/iaas/logging-analytics/doc/overview-logging-analytics.html | https://docs.oracle.com/en/learn/oci-apm-with-opentelemetry/

OCI Application Performance Monitoring (APM)

Distributed tracing and deep visibility into application performance

APM Key Capabilities

- Distributed transaction tracing across microservices and multi-cloud
- End-user monitoring — page load, AJAX calls, service requests, JDBC timing
- Synthetic monitoring — proactive availability checks from global vantage points
- Compatible with OpenTracing, OpenMetrics, and OpenTelemetry (CNCF standards)
- Real User Monitoring (RUM) + backend APM in a single unified view
- Supports WebLogic, Tomcat, TomEE, SpringBoot, JBoss, Jetty, Wildfly

APM Data Types & Signal Sources	
Spans / Traces	Individual service hops in a distributed transaction chain
Metrics	Request rate, error rate, latency (p50, p95, p99 percentiles)
Synthetic	Scripted user journeys run from global vantage points
RUM Sessions	Browser-captured page loads, JS errors, session replays

Source: OCI APM Docs | Oracle A-Team Blog | https://docs.oracle.com/en/learn/oci-apm-with-opentelemetry/ | https://www.ateam-oracle.com/post/end-to-end-monitoring-of-applications-running-on-oracle-cloud-infrastructure-part-3

OCI Operations Insights

ML-driven capacity planning and SQL performance management

What Ops Insights Does

1 Capacity Forecasting

Analyzes 25 months of historical CPU, storage, and memory usage to predict exhaustion 30 days ahead

2 SQL Performance Trends

Identifies common SQL patterns, performance regressions, and tuning opportunities across a database fleet

3 Resource Optimization

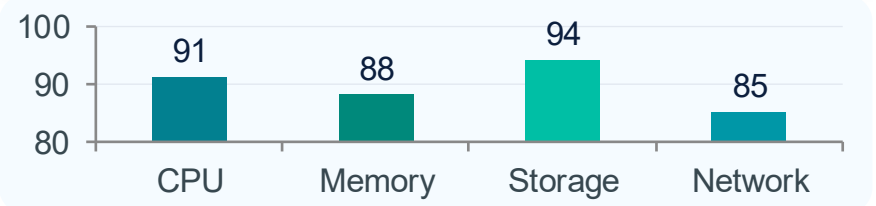
Pinpoints over-provisioned and under-utilized hosts and databases to reduce wasteful spend

4 AWR Analytics

Analyzes Automatic Workload Repository (AWR) data for deep database performance diagnostics & tuning

Business Value Delivered

25 months	Historical data retention for accurate trend forecasting
30 days	Advance warning on capacity shortfalls before they impact SLA
Fleet-wide	Unified SQL tuning insights across all databases in a compartment
News Reports	Weekly automated summaries of utilization highs and inventory changes



Source: OCI Ops Insights Docs | Oracle O&M Platform | https://docs.oracle.com/en-us/iaas/operations-insights/doc/operations-insights.html | https://docs.oracle.com/en-us/iaas/Content/cloud-adoption-framework/monitoring-and-observability.htm

OCI Stack Monitoring & Service Connector Hub

Proactive stack health monitoring and intelligent data orchestration

OCI Stack Monitoring

Full Stack Coverage

Monitor entire application stacks: web servers → app servers → databases → middleware. One unified view.

Proactive Health

Define discovery profiles for known stack types. Auto-discover and monitor components on provision.

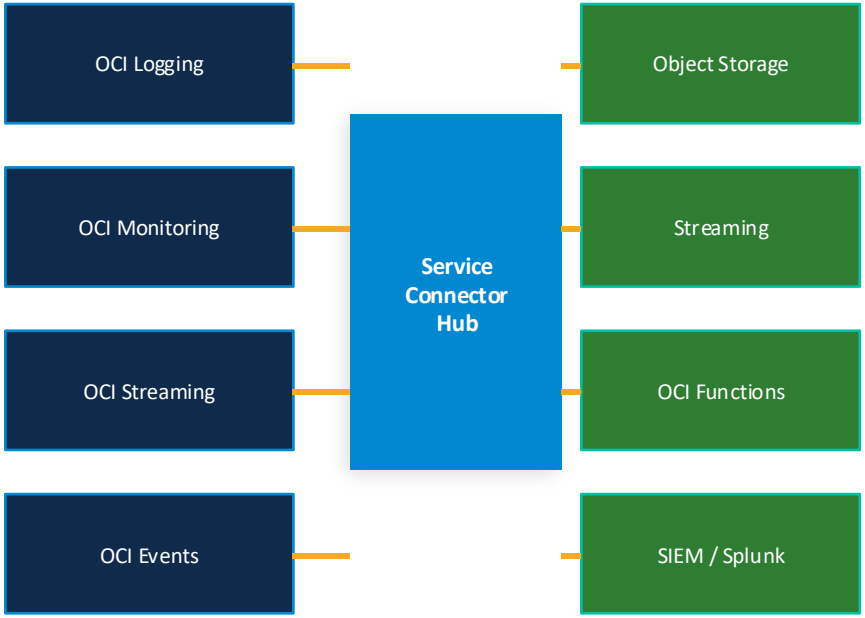
Out-of-Box

Pre-built monitoring for Oracle WebLogic, JBoss, Tomcat, MySQL, PostgreSQL, and OS-level hosts.

Custom Metrics

Publish application-specific custom metrics alongside built-in stack metrics in a unified dashboard.

OCI Service Connector Hub



Free service — no charge for data movement between OCI services via Connector Hub

Source: OCI Stack Monitoring | OCI Service Connector Hub | https://docs.oracle.com/en-us/iaas/Content/cloud-adoption-framework/monitoring-and-observability.htm

OCI Monitoring Service & MQL

Metrics, alarms, and the Monitoring Query Language for intelligent alerting

OCI Monitoring Overview

- Collects service metrics automatically from all OCI resources
- Metrics Explorer provides a unified query interface
- Alarms fire when MQL expressions meet defined thresholds
- Alarm notifications route to OCI Notifications → PagerDuty/Slack
- Custom metrics: push application metrics via REST API or SDK
- Integrates with Grafana for open-source dashboard visualization

MQL Alarm Example

```
CpuUtilization[1m]
  {availabilityDomain = "PHX-AD-1"}
  .groupBy(availabilityDomain)
  .percentile(0.9) > 85
```

Alarm → Action Flow

1

1. Metric Breach

MQL expression evaluates; metric crosses threshold for pendingDuration period

2

2. Alarm Fires

OCI Monitoring transitions alarm to FIRING state with severity (Critical / Warning)

3

3. Notification

Routes to OCI Notifications service — email, HTTPS, Slack, PagerDuty

4

4. OCI Functions

Serverless function triggered to run auto-remediation script or workflow

5

5. Feedback

Resolution outcome logged; Logging Analytics correlates alarm with log events

Source: OCI Monitoring MQL Docs | Oracle A-Team MQL Blog | https://www.ateam-oracle.com/deep-dive-into-oci-observability-and-management-monitoring-query-language-in-oci-alarms

OCI AIOps Implementation Best Practices

Proven design patterns from Oracle's O&M guidance

1 Define Your Observability Strategy First

Map out which OCI O&M services cover each part of your stack before deploying. Avoid tool sprawl by leveraging pre-integrated native services.

2 Prefer Native OCI Services for OCI Workloads

Use OCI Logging Analytics, APM, and Stack Monitoring instead of third-party tools when the majority of workloads run on OCI — they're pre-integrated and cost-optimized.

3 Consolidate Data into Logging Analytics

Route all logs (from OCI Logging, on-prem agents, app servers) into Logging Analytics as the single ML intelligence layer for correlation and anomaly detection.

4 Use Service Connector Hub as Glue

Orchestrate data movement between services (Logging → Analytics → Object Storage → SIEM) using Connector Hub — free, scalable, low-latency pipeline backbone.

5 Govern with Compartments & IAM

Design a dedicated Observability & Management compartment with strict IAM policies. Apply security zones for sensitive log data to prevent unauthorized access.

6 Iterate from Reactive to Proactive

Start with Monitoring + Logging (L1/L2 maturity), then add Logging Analytics ML for L3, then Ops Insights forecasting for L4+ maturity.

OCI AIOps vs Generic AIOps Platforms

Where OCI's native capabilities excel — and where third-party tools fit

Dimension	OCI Native O&M (AIOps)	Third-Party AIOps Platforms
Integration Depth	Native — zero-config for OCI resources	Requires agents and connectors
Log Analytics	ML-powered Logging Analytics built-in	Splunk / ELK — separate licensing
Trace / APM	OCI APM with OpenTelemetry support	Dynatrace, Datadog, New Relic
Capacity Planning	Ops Insights with 25-month ML forecasting	Separate capacity management tools
Data Routing	Service Connector Hub — free, built-in	Kafka / custom pipelines — extra cost
Pricing Model	Consumption-based; no egress charge for O&M	Per-host / per-log licensing — often costly
Multi-Cloud	OCI + on-prem; limited for AWS/Azure native	Fully multi-cloud (Datadog, Dynatrace)
Hybrid Support	Management Agent for on-prem coverage	Strong hybrid via universal agents

End-to-End OCI AIOps Scenario

Tracing an incident from detection to resolution using OCI O&M services

1. **Anomaly Detected** — OCI Monitoring
   OCI Monitoring MQL alarm fires — CPU utilization on app server crosses 90th percentile for 3 min
2. **Log Correlation** — Logging Analytics
   OCI Logging Analytics ML clusters log spikes — detects pattern linked to a recent deployment at 2:15 PM
3. **Trace Analysis** — OCI APM
   OCI APM distributed trace shows payment-service latency spike tied to inventory_db query timeout
4. **Root Cause Identified** — ML Correlation Engine
   Logging Analytics correlates the deployment event + DB query pattern → points to missing DB index
5. **Auto-Remediation** — OCI Functions
   OCI Functions runbook creates DB index automatically; re-routes 20% traffic to healthy replica
6. **Notification & Learn** — Notifications + ML Loop
   OCI Notifications alerts team via Slack; Logging Analytics adds this pattern to the anomaly model

AIOps Architecture Implementation Checklist

Key decisions and components to validate before going live

✓ Data Layer

- All telemetry sources identified and agents deployed
- Data normalization schema defined
- Retention policies configured
- Historical data baseline established (≥90 days)

✓ ML & Analytics

- Anomaly detection models trained on historical data
- Event correlation rules configured and validated
- False positive rate measured and tuned
- RCA workflows tested on known incidents

✓ Automation

- Runbooks documented for top 10 incident types
- Automated playbooks tested in staging environment
- ITSM integration validated (ticket creation/routing)
- Rollback procedures defined for auto-remediation

✓ OCI Specifics

- Compartment structure for O&M defined
- IAM policies for observability services configured
- Service Connector Hub pipelines deployed
- Logging Analytics entities and sources configured

✓ Feedback & Governance

- Model retraining schedule established
- Runbook library reviewed quarterly

Session 2 — Key Takeaways

1. **AIOps architecture is layered —** Data Ingestion → ML Analytics → Automation Engine, connected by a continuous feedback loop.
2. **Three pillars power every platform —** Observe (detect), Engage (analyze), Act (automate) — every AIOps platform maps to this framework.
3. **Data quality is foundational —** ML models are only as good as their inputs — normalization, enrichment, and unified data stores are prerequisites.
4. **Feedback loops drive improvement —** AIOps learns from every resolved incident, continuously improving accuracy, reducing false positives, and expanding automation.
5. **OCI provides full-stack AIOps —** Logging Analytics, APM, Ops Insights, Stack Monitoring, and Connector Hub deliver a native, integrated AIOps platform.
6. **Start with observability, build toward automation —** Deploy monitoring & logging first (L1/L2), then layer in ML analytics (L3), then automation (L4/L5) incrementally.